

Topic 9 - Production & Costs

ECON 101
Sp 23

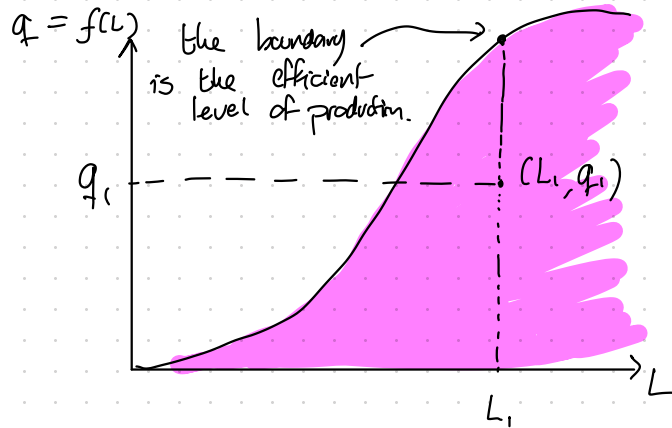


Production Function

Describes what is technically feasible and efficient to produce.

One - Input Production Function

Consider $q = f(L)$, where q is output, L is labor.



Now consider Two-Input production functions.

$$q = F(K, L)$$

$$\text{Average Product of Labor} = AP_L = \frac{q}{L} = \frac{F(K, L)}{L} \quad \left. \vphantom{\frac{F(K, L)}{L}} \right\} \text{ while holding } K \text{ fixed.}$$

$$\text{Marginal Product of Labor} = MP_L = \frac{\Delta q}{\Delta L} = \left(\frac{\partial q}{\partial L} \right)$$

→ If you know what this is

$$AP_K = \frac{q}{K} = \frac{F(K, L)}{K} \quad \left. \vphantom{\frac{F(K, L)}{K}} \right\} \text{ while holding } L \text{ fixed.}$$

$$MP_K = \frac{\Delta q}{\Delta K} = \left(\frac{\partial q}{\partial K} \right)$$

In the one-input technology case, we can derive the AP and MP curves graphically.

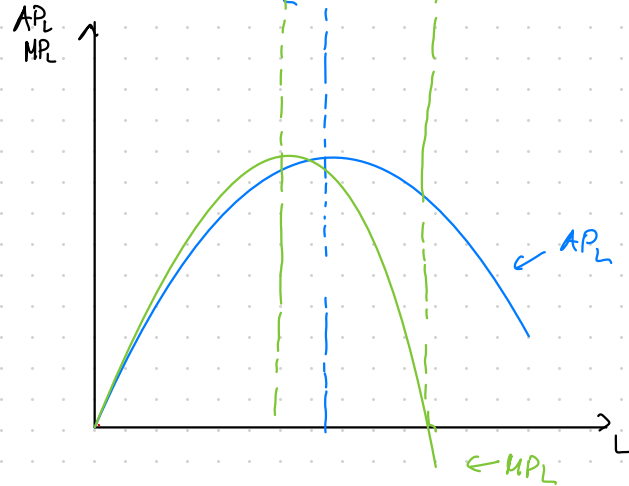
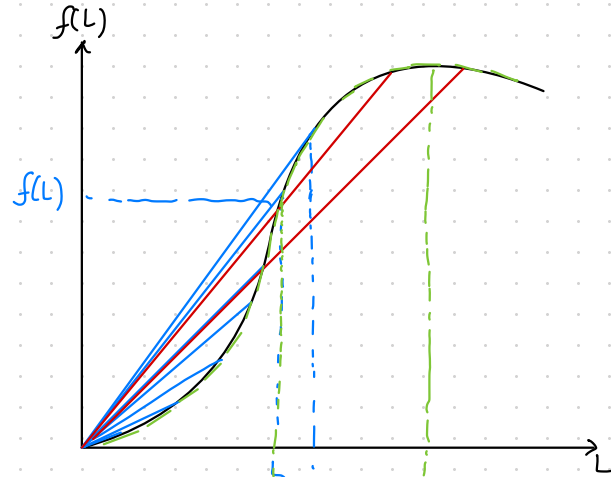
Consider $q = f(L)$

$$MP_L = \frac{\Delta q}{\Delta L} = \frac{\Delta f(L)}{\Delta L} = \left(\frac{df(L)}{dL} \right) \approx \text{slope of } f(L)$$

$$AP_L = \frac{q}{L} = \frac{f(L)}{L}$$

Consider the following secant line through the origin $(0,0)$ and $(L, f(L))$

$L'' \neq L \rightarrow$ slope of this secant: $\frac{f(L) - 0}{L - 0} = \frac{f(L)}{L} = AP_L$



Two-variable Inputs $q = F(K, L)$

An isoquant is the set of input combinations (K, L) that yield the same level of output $\bar{q}$:

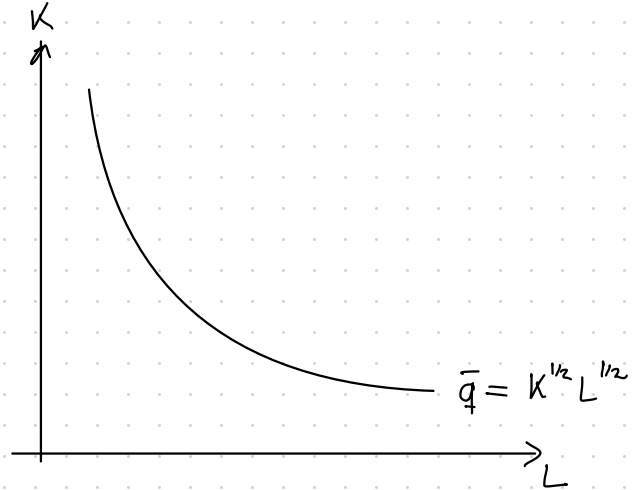
Implicitly it is given: $\bar{q} = F(K, L)$

Ex: $\bar{q} = A K^{1/2} L^{1/2}$

↓

$$\frac{\bar{q}^2}{A^2 L} = \left(\frac{\bar{q}}{A L^{1/2}} \right)^2 = K \quad \text{Isoquant for } \bar{q}$$

- slope of isoquant: $\frac{\Delta K}{\Delta L} = -MRTS = \text{Marginal Rate of Technical Substitution}$
while holding q fixed



$$\Delta q = MP_L \cdot \Delta L + MP_K \cdot \Delta K \quad \Rightarrow \quad \frac{\Delta K}{\Delta L} = - \frac{MP_L}{MP_K} = -MRTS$$

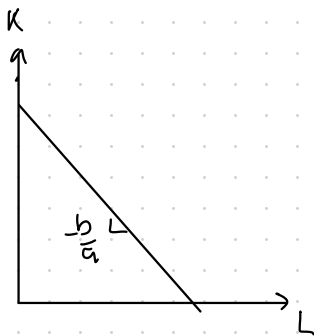
$$MRTS = |\text{slope of isoquant}| = \frac{MP_L}{MP_K}$$

Three Examples:

$$q = aK + bL$$

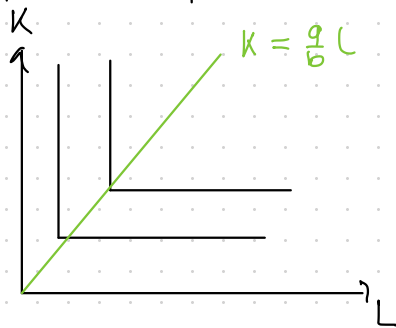
Perfect Substitutes

$$MRTS = \frac{b}{a}$$



$$q = \min \left\{ \frac{K}{a}, \frac{L}{b} \right\}$$

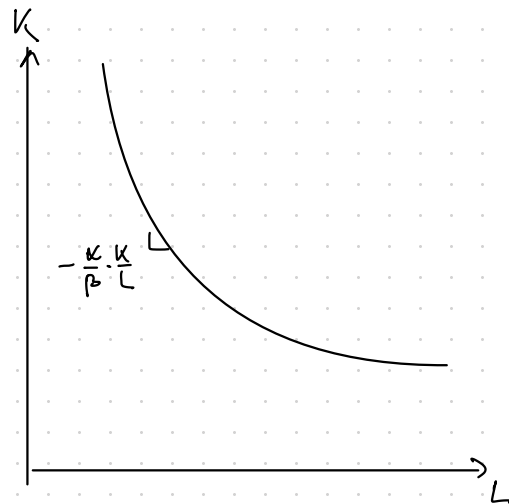
Leontieff production function OR fixed proportions (perfect complements)



$$q = AK^\alpha L^\beta$$

Cobb-Douglas production function

$$MRTS = \frac{\alpha}{\beta} \cdot \frac{K}{L} \quad (L \text{ vs } K)$$



Returns to Scale

One-input production function $q = f(L)$

Increase input by
the constant λ

$$f(\lambda L) = \lambda^s f(L) = \lambda^s q$$

If I am able to
do this, f exhibits some
sort of returns to scale

$$\begin{cases} s = 1 & \text{CRS} \\ s > 1 & \text{IRS} \\ s < 1 & \text{DRS} \end{cases}$$

• $f(L) = L \Rightarrow f(\lambda L) = \lambda L = \lambda f(L) \quad \checkmark$

• $f(L) = L^2 \Rightarrow f(\lambda L) = (\lambda L)^2 = \lambda^2 L^2 = \lambda^2 f(L) \quad \checkmark$

• $f(L) = \sqrt{L} \Rightarrow f(\lambda L) = \sqrt{\lambda L} = \lambda^{1/2} \sqrt{L} = \lambda^{1/2} f(L) \quad \checkmark$

Cost Functions

Production function

$$q = F(K, L)$$

↳ How much q
can I produce given
input combination (K, L)

↑
units of
capital

↑
units
of labor.

← F →
Technology
DUALITY

(Total) Cost Function

$$\text{Cost} = C(q, \text{input prices})$$

How much it costs the firm
to produce q units given
input prices r, w
↑ rental cost of capital
↑ market wage rate

$$\text{Cost} = wL + rK$$

The goal of the firm is to choose optimally its inputs to produce q units of output.

We need to distinguish between:

- Short-Run costs : at least one input is fixed
 - Long-Run costs : all inputs are variable.
-

Short-Run:

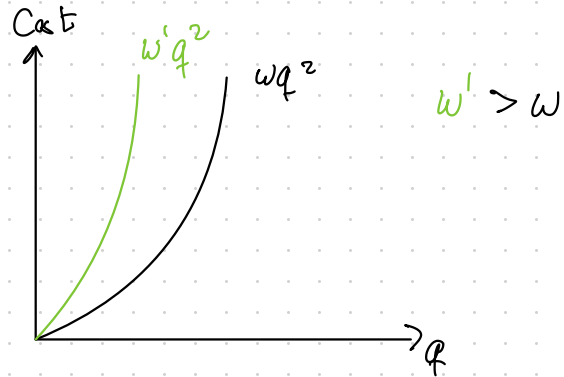
Example: 1-input prod. fn. $q = \sqrt{L}$

To produce q units how many units of labor I need?

$$q = \sqrt{L^*} \Rightarrow L^* = q^2$$

$$\text{Cost} = wL^* = wq^2$$

$$\downarrow$$
$$C(q, w) = wq^2$$



Example : Two-input prod. fn. — Short-Run Costs $q = F(K, L)$

$$q = A K^{1/2} L^{1/2}$$

Short-Run: $K = 100$ units, $r = 1 \rightarrow q = A (100)^{1/2} L^{1/2} = 10A \sqrt{L}$

If the firm wants to produce q units of output, how much labor does it need?

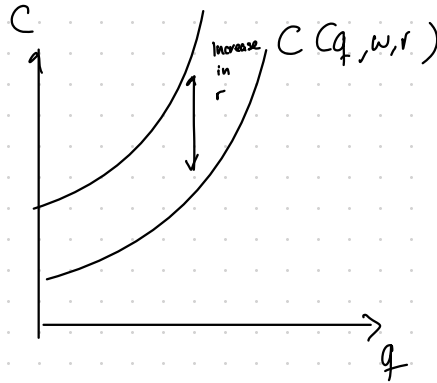
$$\rightarrow q = 10A \sqrt{L^*} \Rightarrow L^* = \left(\frac{q}{10A}\right)^2$$

$$C = wL^* + r\bar{K} = w \left(\frac{q}{10A}\right)^2 + 100$$

$\updownarrow$
 $\nwarrow$ Variable Cost
 $\nearrow$ Fixed Cost

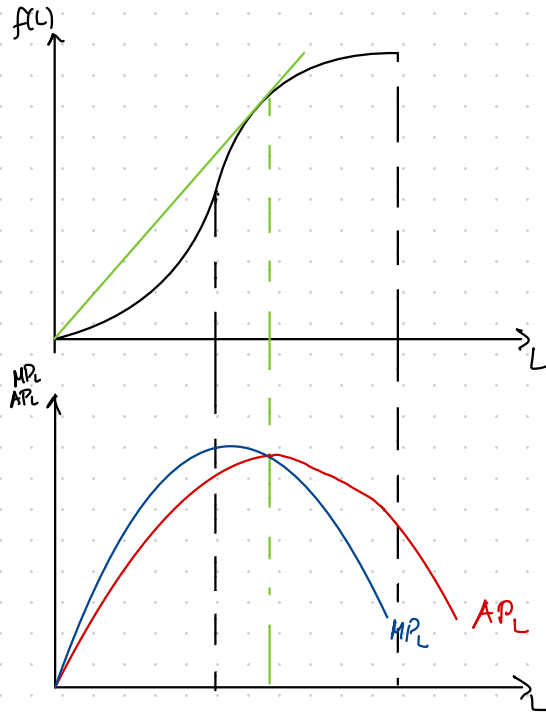
$$C(q, w, r) = \underbrace{\frac{wq^2}{100A^2}}_{VC} + \underbrace{r100}_{FC}$$

$\downarrow$ TC

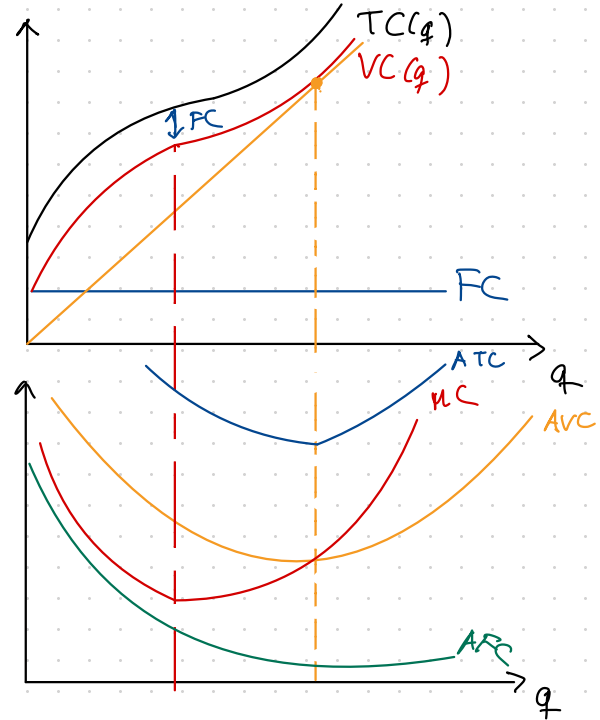


In general The Total Cost (TC) in the Short Run (S)

$$(S)TC = (S)VC + (S)FC$$



DUAL



Example: $q = A K^\alpha L^{1-\alpha}$

Short-Run: $K=100$ $\left. \begin{array}{l} w=1 \\ r=1 \end{array} \right\} \Rightarrow q = 10\sqrt{L}$
 $\alpha = 1/2$
 $A = 1$

$L^* = \frac{q^2}{100} \Rightarrow TC(q) = \frac{q^2}{100} + 100$

$VC(q) = \frac{q^2}{100}$

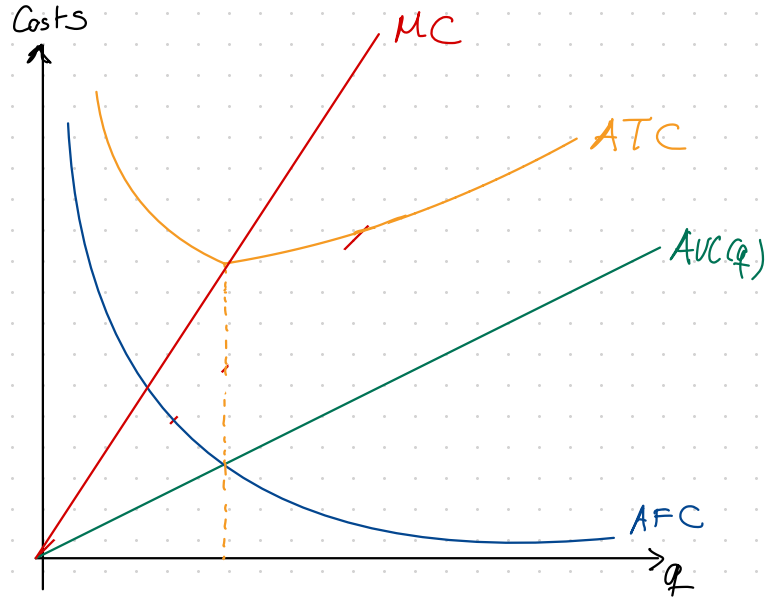
$FC(q) = 100$

$MC(q) = q/50$

$AVC(q) = q/100$

$AFC(q) = 100/q$

$ATC(q) = q/100 + 100/q$



Costs in the Long Run

Goal: choose inputs (all) to minimize cost,

$$C(q, w, r) = wL^* + rK^* = \underset{L, K}{\text{minimize}} \quad wL + rK \quad (\text{Cost Minimization})$$

s.t. $q = F(K, L)$

↓ optimal solutions

Long-Run
cost
function

$L^*(w, r, q)$
 $K^*(w, r, q)$ } Optimal level: Conditional Input demands
of inputs (q)
necessary to
produce q
at prices $w, r,$

How to find the solution?

